

Session 5 — Exercises: Data (collect it well, describe it honestly)

Foundations of Clinical Research · Weekend 3, Saturday · Neudata · #ClearDataClearImpact

Work with your neighbour or group as indicated. No calculator is needed: every exercise is about *choosing* and *interpreting*. Exercises marked **[L2]** use the course dataset FLUID-ICU (simulated) and your group's results pack in `Course/Data/results_packs/`. Exercise 8 becomes part of your proposal (Part 3).

#	Exercise	When	Time
1	Classify the variables	5.1 pair task	3 min
2	Fix the bad CRF items	5.1 pair task	3 min
3	Choose the right summary	5.3 check / homework	3 min
4	Judge a bad figure	homework	5 min
5	[L2] Spot the problems in the raw FLUID-ICU data	homework (after the 5.2 demo)	5 min
6	[L2] Read your group's Table 1	5.4 group worksheet	10 min
7	[L2] Your group's question	5.2 group task	4 min
8	Proposal builder 5 worksheet	proposal builder	30 min

Exercise 1 — Classify the variables

For each variable, tick **one** type.

#	Variable (ICU study of septic shock)	Continuous	Count	Binary	Nominal	Ordinal	Time-to-event
a	Age at ICU admission (years)	☐	☐	☐	☐	☐	☐
b	Sex (male / female)	☐	☐	☐	☐	☐	☐
c	Glasgow Coma Scale on admission (3–15)	☐	☐	☐	☐	☐	☐
d	Main source of sepsis (lung / urine / abdomen / skin / other)	☐	☐	☐	☐	☐	☐

#	Variable (ICU study of septic shock)	Continuous	Count	Binary	Nominal	Ordinal	Time-to-event
e	Number of vasopressors running at 24 h (0, 1, 2, 3)	☐	☐	☐	☐	☐	☐
f	Cumulative fluid balance at day 5 (ml/kg)	☐	☐	☐	☐	☐	☐
g	Alive at day 28 (yes / no)	☐	☐	☐	☐	☐	☐
h	Time from ICU admission to death (patients alive at day 28 are censored)	☐	☐	☐	☐	☐	☐
i	Pain: none / mild / moderate / severe	☐	☐	☐	☐	☐	☐
j	Blood group (A, B, AB, O)	☐	☐	☐	☐	☐	☐

Bonus: which ONE of these would you most like to collect as an exact number rather than in groups, and why?

Exercise 2 — Fix the bad CRF items

Each item below can be filled in two different ways by two different nurses. Rewrite it so that it has **only one possible reading**. Check yourself: unit? time point? one answer only? an "unknown / not done" option?

#	Bad item	What is wrong?	Your rewritten item
a	Fluid given: _____		
b	Weight: _____		
c	Comorbidities: _____ _____		
d	Lactate high? Y / N		

Exercise 3 — Choose the right summary

For each variable, choose the summary you would report in Table 1 and the best graph.

#	Variable (200 ICU patients)	Shape / type	Summary	Graph
a	Age (years), bell-shaped			
b	ICU length of stay (days): most 1–5 days, a few > 30 days			
c	Died in hospital (yes / no)			
d	C-reactive protein (mg/L): long right tail			
e	Source of sepsis (5 categories)			
f	Systolic blood pressure, compared between two treatment groups			

Exercise 4 — Judge a bad figure

A poster in the hospital corridor shows this figure (fluid balance at day 5, 24 patients per group):

![Dynamite plot](../Resources/figures/en/sat3/dynamite_bad.png)

1. What does each bar show, and what does the error bar show? What is missing?
2. Name **two things the reader cannot see** in this figure.
3. Redraw it (sketch) the way you would show it in a paper. What do you gain?

A second poster shows ward mortality as bars of 31% (Ward A) and 29% (Ward B), with the y-axis running from 28% to 32%.

4. What impression does it give, and how would you fix it?
5. A third poster puts hand-hygiene compliance (%) and infection rate on one graph with two different y-axes and writes "hygiene works!". Give two reasons to be careful.

Exercise 5 — [L2] Spot the problems in the raw FLUID-ICU data

Below are six cells copied from the raw file `Course/Data/fluid_icu_raw.csv` (as the four ICUs sent it). For each, say what is wrong and what the cleaning script should do.

#	Column	Value in the raw file	What is wrong?	What should the script do?
a	sex	m, male, Female		
b	admission date	14/03/2023 in one row, 14-Mar-2023 in another		

#	Column	Value in the raw file	What is wrong?	What should the script do?
c	lactate (ICU C)	23 . 4		
d	lactate	999		
e	age	210		
f	patient ID	FLU-139 appears in two identical rows		

Bonus: why must the raw file never be edited by hand? Name two things the cleaning log records for every change.

Exercise 6 — [L2] Read your group's Table 1 (results pack)

Open your group's results pack (Course/Data/results_packs/Results_Pack_Qn.docx, Table 1: baseline characteristics by fluid-balance group).

1. How many patients are in each column? Why is the overall number 618 and not 620?
2. For **three** variables, write the summary used (mean (SD), median (IQR) or n (%)) and why.

Variable	Summary used	Why this summary?

3. Were the two groups alike at the start? Name the **biggest** difference between the negative- and positive-balance groups.
4. Which variable has the most missing values, and how many?
5. Write **two sentences** describing the patients, as they would appear at the start of your Results section (numbers in brackets, no p-values).

> _____

6. Looking at this table, would you expect the crude association between positive fluid balance and death to be larger or smaller than the adjusted one? Why?

Exercise 7 — [L2] Your group's question (5.2)

	Your answer
Group question (Q1–Q5), copied from page 1 of your results pack	
P — population	
E — exposure (variable name in the data dictionary) and its type	
C — comparator	
O — outcome (variable name) and its type	
One-sentence research question	
Which test will compare the groups? Which model will adjust? (guess now, check in Session 6)	

Exercise 8 — Proposal builder 5 worksheet (group, 30 minutes)

Write your answers here first, then copy them into Part 3 of the group proposal template. (Your **own** question, not the course dataset.)

8.1 Variables table (8 min)

Role	Variable	How measured (unit, time point)	Type
Primary outcome			
Secondary outcome (optional)			
Exposure / intervention			
Confounder 1			
Confounder 2			
Confounder 3			
Descriptive (e.g. age, sex)			

Tip: if your design is an RCT, confounders are less of a problem (randomisation), but list the key baseline variables you will show in Table 1.

8.2 Mini-CRF: 6–10 items (10 min)

#	Question exactly as the data collector will read it	Answer format (boxes / ticks / unit)	"Unknown / not done" option?
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐

8.3 Data dictionary: 3–5 rows + one data-quality rule (6 min)

Variable name	Label (meaning)	Type	Unit / codes	Allowed values

Data-quality plan (one or two sentences): who checks the forms, how often, and what range/consistency check you will run.

> _____

8.4 Data-management plan (6 min)

Question	Our answer
Paper CRF with double data entry, or electronic data capture (e.g. REDCap / KoBoToolbox)?	

Question	Our answer
Coded study IDs? Where is the linking log kept, and who can open it?	
Validation rules at entry (two examples from your data dictionary)	
Audit trail / cleaning log: who records changes?	
Backups (3-2-1) and data-sharing statement in the consent form	